

EMAIL chrislpkuo@berkeley.edu

INTERESTS • Microlocal Sheaf Theory
 • D-modules
 • Geometric Representation Theory
 • Symplectic and Contact Geometry
 • Noncommutative geometry
 • Mirror symmetry

EDUCATION AND TRAINING • Ph.D. in Mathematics, UC Berkeley, 2016 - 2022, awarded on May 13, 2022.
 Advisor: Vivek Shende.
 Thesis title: Symplectic geometric methods in microlocal sheaf theory.
 • B.S. in Mathematics, National Taiwan University, Taipei, Taiwan, 2010 - 2014.
 Advisor: Chung-Jun Tsai.
 Thesis title: A calculation for the Hard Lefschetz map on six-manifolds constructed by Coisotropic Luttinger Surgery.

APPOINTMENTS • Postdoc at the Max Planck Institute for Mathematics, Summer 2025 - Present.
 • Postdoc for Noncommutative Algebraic Geometry, SLMath, Spring 2024.
 • Assistant Professor (RTPC) of Mathematics, University of Southern California, Fall 2022 - Spring 2025.
 — Herbert and Ruth Busemann Assistant Professor of Mathematics, Fall 2023 - Spring 2024.

RESEARCH • Tomohiro Asano, Yuichi Ike, C. Kuo, and Wenyuan Li. *Sheaf quantization of non-exact Lagrangians and applications*, in preparation.
 • Sheel Ganatra, C. Kuo, Wenyuan Li, and Haosen Wu. *A microlocal criterion for compactness preserving integral kernels*, in preparation.
 • Tomohiro Asano, Yuichi Ike, C. Kuo, and Wenyuan Li. *C^0 -rigidity of Legendrians and coisotropics via sheaf quantization*, [arXiv:2510.01746](https://arxiv.org/abs/2510.01746).
 • C. Kuo and Harold Williams. *Tropical Lagrangian coamoebae and free resolutions*. [arXiv:2412.18071](https://arxiv.org/abs/2412.18071).
 • C. Kuo and Wenyuan Li. *Relative Calabi-Yau structure on microlocalization*. [arXiv:2408.04085](https://arxiv.org/abs/2408.04085), submitted in August 2025.
 • C. Kuo and Wenyuan Li. *Duality and Kernels in Microlocal Geometry*, IMRN 2025 (2025): rna070. ([arXiv:2405.15211](https://arxiv.org/abs/2405.15211))
 • Laurent Côté, C. Kuo, David Nadler, and Vivek Shende. *The microlocal Riemann-Hilbert correspondence for complex contact manifolds*. [arXiv:2406.16222](https://arxiv.org/abs/2406.16222), submitted in April 2025.
 • C. Kuo, Vivek Shende, and Bingyu Zhang. *On the Hochschild cohomology of Tamarkin categories*. [arXiv:2312.11447](https://arxiv.org/abs/2312.11447), submitted in June 2025.
 • C. Kuo and Wenyuan Li. *Spherical adjunction and Serre functor from microlocalization*. [arXiv:2210.06643](https://arxiv.org/abs/2210.06643), submitted in February 2025.
 • Laurent Côté, C. Kuo, David Nadler, and Vivek Shende. *Perverse Mi-*

crosheaves. arXiv:2209.12998, to appear in Compositio.

- C. Kuo. *Wrapped sheaves*, Advances in Mathematics 415 (2023): 108882. (arXiv:2102.06791)

PH.D. STUDENTS
SUPERVISED

- Haosen Wu, USC, co-supervised with Wenyuan Li under Sheel Ganatra.

TEACHING
EXPERIENCE

- Instructor, V5D4, Introduction to Microlocal Sheaf Theory, University of Bonn, Summer 2026, notes available at <https://chriskuo0.github.io/homepage/V5D4-ItMST/>
- Instructor, Math 125, Calculus I, University of Southern California, Spring 2025
- Instructor, Math 425a, Fundamental Concepts of Analysis, University of Southern California, Spring 2025
- Instructor, Math 425a, Fundamental Concepts of Analysis, University of Southern California, Fall 2024
- Instructor, Math 144, Foundations of Statistics, University of Southern California, Fall 2023
- Instructor, Math 245, Mathematics of Physics & Engineering I, University of Southern California, Spring 2023
- Instructor, Math 125g, Calculus I, University of Southern California, Fall 2022
- Graduate Student Instructor, Math 110, Linear Algebra, UC Berkeley, Spring 2020
- Graduate Student Instructor, Math 16B, Analytic Geometry and Calculus, UC Berkeley, Spring 2019
- Graduate Student Instructor, Math 54, Linear Algebra and Differential Equations, UC Berkeley, Fall 2017 and Fall 2018
- Graduate Student Instructor, Math 53, Multivariable Calculus, UC Berkeley, Spring 2017
- Graduate Student Instructor, Math 1A, Calculus, UC Berkeley, Fall 2016
- Teaching Assistant, Geometry (Required course for juniors), National Taiwan University, Fall 2013

SEMINARS

- Organizer, Bonn Symplectic Geometry Seminar, University of Bonn, Fall 2025 - Present
- Organizer, Microlocal learning seminar, University of Southern California, Spring 2025
- Co-organizer, Topology seminar, University of Southern California, Fall 2022 - Spring 2025
- Organizer, Deformation quantization modules learning seminar, Syddansk Universitet, Fall 2021 - Spring 2022

INVITED TALKS

- Characteristic cycles on contact manifolds, Topology Seminar, USC, August 2026.
- Perverse microsheaves on contact manifolds, seminar at the Chair of Arithmetic Geometry, EPFL, November 2025.
- Riemann-Hilbert on contact manifolds, MPI-Oberseminar, MPIM, August 2025.

- Tropical Lagrangian coamoebae and free resolutions, Syzygies and Mirror Symmetry Virtual Seminar, April 2025.
- Symplectic Geometry and Sheaves, Geometry and Topology Seminar, UC Irvine, November 2024.
- A quick introduction of microlocal sheaf theory, NAG GRT Seminar, SLMath, February 2024.
- Globalizing the microlocal Riemann-Hilbert correspondence, Algebraic Geometry Seminar, UC Davis, January 2024.
- Globalizing the microlocal Riemann-Hilbert correspondence, Algebraic Geometry Seminar, Academia Sinica, January 2024.
- Globalizing the microlocal Riemann-Hilbert correspondence, Geometry Seminar, Kyushu University, January 2024.
- Microlocal Riemann-Hilbert correspondence, QM Research Seminar, University of Southern Denmark, December 2023.
- Microlocal Riemann-Hilbert correspondence, Bonn symplectic geometry seminar, December 2023.
- Perverse Microsheaves, Mathematics Seminars, Montana State University, October 2023.
- Perverse Microsheaves, Differential geometry and symplectic topology seminar, University of Minnesota, September 2023.
- Sheaf Theoretic Methods in Symplectic Geometry, NCTS Differential Geometry Seminar, National Center for Theoretical Sciences, Taipei City, May 2023.
- Microlocal sheaf theory in noncommutative geometry, Algebraic Geometry Seminar, Academia Sinica, Taipei City, May 2023.
- Microlocal sheaf theory in noncommutative geometry, UCSB Seminar on Geometry and Arithmetic, University of California, Santa Barbara, May 2023.
- Wrapped sheaves, Topology Seminar in USC, September 2022
- Wrapped sheaves, QM Research Seminar in SDU, March 2022
- Wrapped sheaves, U. Iowa Topology Seminar, December 2021

SEMINAR AND TALKS

- Recollections on analytic stacks, Arbeitsgruppe Arithmetische Geometrie und Darstellungstheorie (ARGOS), MPIM, Fall 2025.
- Sheaf quantization and the topology of cotangent bundles, guest lecture in Sheel Ganatra's Math 699, USC, Fall 2022
- Nondisplaceability results for Lagrangians in toric symplectic manifolds, Southern California Symplectothon, October 2022
- Hochschild homology, Deformation quantization modules learning seminar, Syddansk Universitet, February 2022
- Kernels, Deformation quantization modules learning seminar, Syddansk Universitet, December 2021
- Algebroids, Deformation quantization modules learning seminar, Syddansk Universitet, November 2021
- Nonlinear regularity and local existence, J-Holomorphic curves learning seminar, Syddansk Universitet, October 2021
- Deformation of a sheaf of rings, Deformation quantization modules learning seminar, Syddansk Universitet, October 2021

- E_n -algebras and iterated loop spaces, Geometric Representation Theory seminar, UC Berkeley, Spring 2020
- Speaker, Wrapped Floer Homology Learning Seminar, Fall 2019
- Hochschild-Kostant-Rosenberg theorem through loop spaces, Geometric Representation Theory seminar, UC Berkeley, November 27 and December 14, 2018
- Koszul duality for algebras, Deformation Theory Seminar, UC Berkeley, October 15 2017
- Gerbes. Mayer-Vietoris calculations, K-theory Seminar, UC Berkeley, April 2017
- Adams operation, K-theory Seminar, UC Berkeley, February 2017
- Winter Camp for Fundamentals in Mathematics for College Students, National Taiwan University, January 23-February 8, 2013

PROFESSIONAL SERVICE

- Beisitzer
 - Introduction to symplectic geometry, University of Bonn, Summer 2025
 - Course on Locally analytic representations and geometric representation theory, University of Bonn, Summer 2026
- Master's thesis second reader, University of Bonn, Summer 2025
- Topology Screening Exam Committee, University of Southern California, Fall 2024
- Co-organizer, The Los Angeles Workshop on Representations and Geometry (LAWRGe), Summer 2023
- Referee for Annals of Mathematics, Compositio, Geometry & Topology, Selecta, and Journal of Topology ($\times 2$).

AWARDS AND HONORS

- Shing-Tung Yau College Student Mathematics Contest, Bronze medal in geometry and topology, 2014
- Honors Program in Mathematics, National Taiwan University, 2014
- Chow Hung-Ching Scholarship, Institute of Mathematics, Academia Sinica, 2013